

Anantha Narayanan R

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EDUCATION

Sri Sairam Engineering College

Bachelor of Technology in Artificial Intelligence and Data Science; CGPA: 8.5/10

Chennai, Tamil Nadu

Nov. 2021 – Apr. 2025

EXPERIENCE

Product Engineer

June 2025 – Present

Oraczen

Hyderabad, India

- Engineered an **advanced Text-to-SQL chatbot** by refactoring a LangGraph application into a **ReAct Agent architecture**, cutting latency by 40% and raising decision-making accuracy from 40% to 90%
- Designed a **context off-loading mechanism** to prevent context degradation in long-running agent sessions, achieving 95% accuracy and a 99% reliability score across extended multi-turn interactions
- Built an **LLM-as-a-Judge evaluation framework** for continuous quality monitoring across production AI systems
- Developed a **Deep-Researcher Multi-Agent System** where planner and researcher agents generate reports via web search, Text-to-SQL analysis, and image generation (Nano-Banana), layering a **LongWriter-style plan-then-write pipeline** for coherent long-form output, lifting accuracy from 35% to 90% and automating 80% of manual research
- Delivered real-time **voice agents** over **WebRTC** with tool-calling support, extracting **knowledge graphs**, actionable items, and summaries from calls, with chat over voice logs via **Qdrant**-backed long-term memory
- Established a **prompt orchestration layer** using **XML tag-based structured prompting** across the Deep-Researcher and voice agents, isolating untrusted input to harden against **prompt injection**, and added **prompt-level guardrails** to keep agent behaviour on-policy
- Built core **Zen modules** for the company's **Zen Platform**, including **ZenTools** and **ContextZen**

Software Engineer

July 2024 – June 2025

Launch Ventures

Hyderabad, India

- Built **Wisdom Chat**, a RAG application delivering personalized guidance grounded in Bhagavad Gita texts, using a **HyDE-based RAG pipeline** with LlamaIndex, NeMo Guardrails, and RAGAS-driven quality evaluation
- Shipped a Flux-powered on-demand image generation feature that visualises abstract spiritual concepts for users
- Engineered the backend for a **Market Research Application** in **LangGraph** with multi-level node classification and validation steps, running web research across **Exa**, **Tavily**, and **News API** and exporting findings as structured **CSV** reports
- Implemented a **plan-then-write pipeline** for report generation, drafting sections independently for coherent long-form output
- Developed a production-grade async pipeline with **FastAPI**, **Celery**, **Redis**, and **Postgres**-backed **checkpointing**, scheduling **cron**-driven daily runs that email each morning's research report as a **CSV** attachment

PROJECTS

Promptly: AI Prompt Optimization Platform | [GitHub](#) | *Personal*

Apr. – Jun. 2026

- Built a full-stack SaaS platform that optimizes LLM prompts through a multi-agent **“council” pipeline**: four models rewrite a prompt, blind peer-review each other's work, and a chairman model synthesizes the result
- Orchestrated the pipeline with **LangGraph** (conditional routing, Postgres-backed checkpointing) and decoupled long-running LLM calls using **FastAPI**, **Celery**, and **Redis**, streaming live progress to the client over SSE
- Architected the backend as a modular monolith with dual authentication (Supabase JWT + API keys), credit-based usage metering, and Postgres RLS, paired with a **Next.js 14** job-tracking and analytics dashboard

DebateMind: AI Debate Coaching Platform | [GitHub](#) | *Personal*

2026

- Built an AI debate-coaching application for the **WeMakeDevs × Cognition Hackathon**, featuring persistent per-user **knowledge-graph memory** that tracks argument history, reasoning patterns, and improvement areas across sessions
- Designed a multi-agent coaching pipeline with **LangGraph**, using **Cognition** to build and query per-user knowledge graphs backed by **Neo4j** and **pgvector** for long-term semantic and relational memory
- Implemented asynchronous task orchestration with **Celery** and **Redis** behind a **FastAPI** backend, enabling scalable, low-latency sessions with real-time feedback via **OpenRouter**

TECHNICAL SKILLS

Languages: Python, C/C++, SQL, HTML/CSS, JavaScript, TypeScript

AI/ML Frameworks: LangChain, LangGraph, LlamaIndex, CrewAI, Google ADK, Transformers, OpenAI API, Groq

Web Frameworks & Protocols: Vue.js, FastAPI, Tailwind CSS, Pinia, MCP, A2A, AG-UI, WebRTC

Databases & Vector Stores: PostgreSQL, ClickHouse, MySQL, Pinecone, ChromaDB, Qdrant, MongoDB, Neo4j

Developer Tools: Git/GitHub, Docker, Dev Containers, Cursor, Antigravity, pytest

Libraries: pandas, NumPy, Matplotlib, RAGAS, NeMo Guardrails, Custom RAG Pipelines